

SURAKSHAMAP AI

Product Requirements Document

Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

AI + GIS-based Disaster Risk Assessment and Decision Support Platform

**Core product chain: Hazard Detection -> Exposure -> Vulnerability -> Risk -> Capacity -> Relocation
Priority -> Safe Destination**

Prepared as a project / hackathon-ready PRD

1. Executive Overview

SurakshaMap AI is a decision-support platform designed to help disaster-management authorities identify vulnerable habitations, quantify multi-hazard risk, assess carrying capacity, and prioritize immediate relocation needs. The platform combines geospatial information, hazard indicators, population exposure, vulnerability factors, infrastructure data, shelter capacity, and field observations into an explainable risk and relocation workflow.

Output	What the system answers
Risk Zone	Where are the red/orange/yellow/safe assessed areas?
Exposure	How many people and critical assets are affected?
Vulnerability	Which communities are less able to withstand or evacuate from the hazard?
Capacity	Can shelters / safer areas accommodate the affected population?
Relocation Need	Which habitations require the most urgent authority attention?
Destination Options	Which candidate locations have comparatively lower assessed risk and available capacity?

2. Problem Statement

Vulnerable habitations can be exposed to floods, landslides, extreme rainfall, river erosion, earthquakes, cyclones, heat, wildfire and other hazards. Authorities often need to combine maps, demographic information, infrastructure records, weather observations, shelter capacity and field reports during time-sensitive decisions. When these inputs are fragmented, it becomes difficult to quickly determine which habitations are most at risk, how many people may require assistance, whether available facilities can absorb displaced populations, and which relocation options should be reviewed first.

The product addresses this gap by creating a single, geospatial, explainable decision-support workflow that turns heterogeneous data into operationally useful assessments.

3. Product Vision & Goals

Vision

Move disaster response from fragmented reactive assessment toward faster, evidence-based preventive planning and coordinated relocation readiness.

Primary Goals

- Identify hazard-prone habitation clusters.
- Generate dynamic assessed risk zones.
- Estimate population exposure and vulnerability.
- Assess shelter and relocation-area carrying capacity.
- Identify capacity deficits.
- Prioritize habitation-level relocation needs for authority review.
- Surface potential safer destinations and feasible routes.
- Provide explainable evidence, confidence/validation status, timestamps and data provenance.

Non-Goals

- Replace trained disaster-management officials.

- Automatically issue or execute forced evacuation orders.
- Guarantee that any location is completely safe.
- Provide detailed structural engineering certification.
- Make autonomous life-or-death decisions without human review.

4. Users & Use Cases

User	Primary need	Key actions
District Disaster Management Authority	Rapid district-wide prioritization	Review red zones, capacity deficits, relocation priorities and reports
State Disaster Management Authority	Cross-region monitoring	Compare areas, aggregate impact and track preparedness
Local Administration / Panchayat	Habitation-level action	Verify field conditions, shelters, routes and population information
Emergency Response Teams	Operational awareness	Check affected areas, access routes, shelters and critical assets
Analyst / Planner	Planning and validation	Study trends, model performance and historical risk
Field Officer	Ground truth	Submit GPS-tagged observations, photos and shelter updates

Representative Use Cases

- A sudden rise in rainfall and river level causes a habitation to move from orange to red assessed risk.
- The system detects that expected displaced population exceeds available shelter capacity.
- A primary evacuation road is flagged as hazardous and an alternate route is shown.
- Officials inspect why a habitation is marked critical before taking action.

5. Core User Journey

DATA COLLECTION -> VALIDATION -> HAZARD ANALYSIS -> EXPOSURE/VULNERABILITY -> RISK -> CAPACITY -> RELOCATION PRIORITY -> AUTHORITY REVIEW

Example: heavy rainfall increases flood indicators around a village. The system combines rainfall, river/water levels, terrain, population, accessibility and shelter capacity; recalculates the assessed risk; estimates the exposed population and shelter deficit; then surfaces candidate destinations for authority review.

6. Product Scope & Features

6.1 Interactive GIS Risk Map

- Administrative boundaries and habitation locations
- Hazard layers and risk zones
- Population exposure
- Roads, bridges and critical infrastructure
- Shelters and available capacity
- Evacuation / relocation routes
- Filters by hazard, geography, population, risk and relocation priority

6.2 Hazard Detection & Multi-Hazard Analysis

Initial MVP hazards: flood, landslide, extreme rainfall and river erosion. Future versions can add earthquake, cyclone, wildfire, heat and drought.

6.3 Vulnerability Assessment

- Population density and household concentration
- Children, older adults and persons with disabilities, where reliable aggregated data is legally and operationally available
- Housing condition and access to services
- Distance to shelters and healthcare
- Road and communication accessibility
- Historical disaster impact

6.4 Capacity Assessment

- Maximum safe capacity
- Current occupancy
- Usable floor area
- Water and sanitation availability
- Food / emergency-supply support
- Medical support
- Accessibility and transport access

6.5 Alerts

Generate configurable alerts when a habitation changes risk category, exposure exceeds thresholds, shelter capacity becomes insufficient, routes become unavailable, or field evidence indicates worsening conditions.

6.6 Field Mobile Interface

- GPS-tagged reports
- Photo upload
- Affected-population updates
- Road blockage reports
- Shelter occupancy updates
- Verification workflow

7. Risk & Capacity Methodology

7.1 Risk Model

$$\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability}$$

The implementation should normalize inputs and document the chosen weights. Thresholds for green/yellow/orange/red should be configurable and validated for the selected geography rather than treated as universal standards.

Score	Assessed zone	Interpretation
0-25	Green	Relatively lower assessed risk
26-50	Yellow	Moderate assessed risk
51-75	Orange	High assessed risk
76-100	Red	Very high assessed risk

7.2 Carrying Capacity

Available Capacity = Maximum Safe Capacity - Current Occupancy

Capacity deficit = Population Requiring Relocation - Available Capacity. A positive deficit indicates that additional capacity or a different relocation plan needs authority review.

7.3 Relocation Priority

A normalized priority score can combine assessed risk, exposed population, vulnerability, capacity deficit and route accessibility. The result should be presented with the factors that contributed to the score, plus a validation status and timestamp.

8. AI / ML Requirements

- Risk prediction using current and historical hazard indicators.
- Hotspot detection for clusters of vulnerable habitations.
- Population impact estimation.
- Anomaly detection for sensor/weather streams.
- Decision-support ranking of relocation priorities using configurable factors.
- Optional remote-sensing / computer-vision analysis for flood extent, road blockage or damage detection.

Explainability Requirement

AI output must be interpretable. The interface should show the main contributing factors instead of only a single model score. Example: “High priority because flood exposure is high, exposed population is large, shelter capacity is insufficient and route accessibility is poor.”

9. Data Requirements

Data category	Examples	Typical use
Geographic	Administrative boundaries, habitation points, roads, rivers, elevation, land use	Spatial context and route analysis
Hazard	Flood extent, rainfall, river levels, landslide susceptibility, erosion zones, historical events	Hazard intensity / probability
Population	Population, density, aggregated vulnerable-group indicators, housing	Exposure and vulnerability
Infrastructure	Schools, hospitals, shelters, bridges, water, electricity	Critical asset exposure and capacity
Real-time	Sensors, weather observations, satellite-derived observations, field reports	Dynamic updates
Operational	Shelter occupancy, supplies, route status, response resources	Relocation and response planning

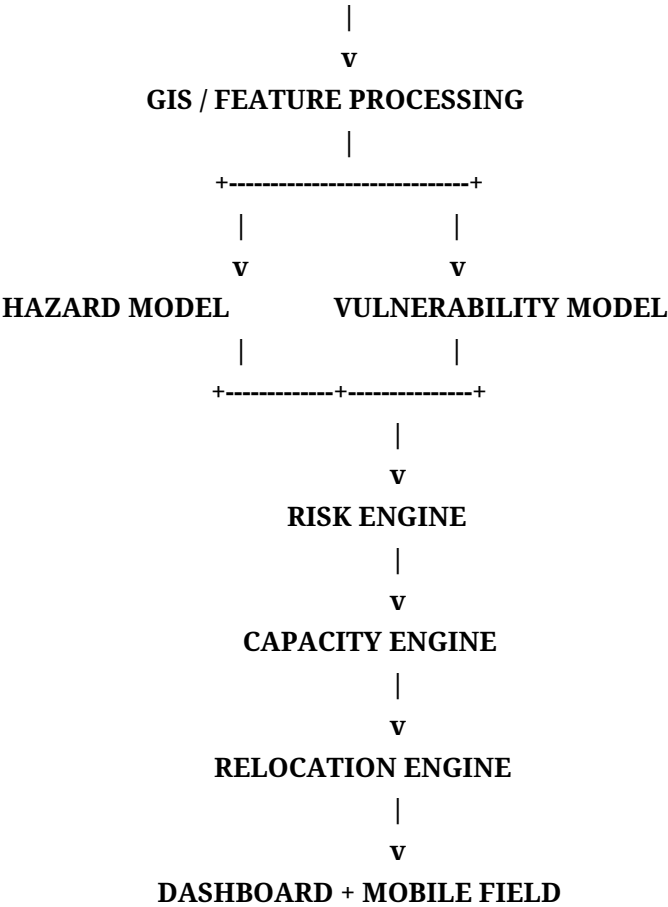
10. System Architecture

DATA SOURCES

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INGESTION + VALIDATION



Suggested Technology Stack

Layer	Suggested options
Frontend	React / Next.js, Tailwind CSS, Leaflet or Mapbox GL
Backend	Python FastAPI or Node.js REST APIs
Database	PostgreSQL + PostGIS
AI/ML	Python, Pandas, Scikit-learn, XGBoost/LightGBM; PyTorch/TensorFlow where justified
GIS processing	GeoPandas, Rasterio, GDAL, PostGIS
Deployment	Cloud platform, object storage, containers, monitoring

11. Dashboard & User Experience

Executive Dashboard

Metric	Example display
Red zones	27
Orange zones	61
Population at risk	48,620
Critical relocation cases	9
Available shelter capacity	31,450
Capacity deficit	8,720

Habitation Detail Page

- Basic information: name, administrative location, population, area

- Hazard panel: flood, landslide, erosion and other relevant risks
- Vulnerability panel: demographic and infrastructure indicators
- Capacity panel: shelter capacity, occupancy and deficit
- Relocation panel: priority level and candidate destinations
- Evidence panel: main factors, source, timestamp and validation status

12. Functional Requirements

ID	Requirement	Description
FR-01	Map Visualization	Display hazard and habitation information on an interactive GIS map.
FR-02	Risk Classification	Assign configurable risk categories to monitored habitations.
FR-03	Multi-Hazard Analysis	Combine multiple applicable hazards into a habitation-level assessment.
FR-04	Population Exposure	Estimate population exposed to identified hazards.
FR-05	Vulnerability Analysis	Calculate vulnerability from demographic, infrastructure and accessibility indicators.
FR-06	Capacity Assessment	Calculate shelter and relocation-area carrying capacity.
FR-07	Capacity Deficit	Identify where relocation demand exceeds available capacity.
FR-08	Relocation Priority	Calculate and display relocation priority with contributing factors.
FR-09	Safe Destination Options	Surface candidate lower-risk locations with available capacity.
FR-10	Route Analysis	Display feasible evacuation / relocation routes when route data is available.
FR-11	Alerts	Generate configurable alerts for critical changes.
FR-12	Field Reports	Allow authorized field officers to submit and verify observations.
FR-13	Explainability	Show reasons, evidence, timestamp and validation status for assessments.
FR-14	Reports	Generate downloadable assessment reports.

13. Non-Functional Requirements

Area	Requirement
Performance	Responsive map/API behavior under normal operating conditions.
Scalability	Support expansion from pilot geography to district/state scope.
Availability	Designed for resilient operation during high-demand emergency periods.
Security	Role-based access control, secure APIs, encryption and audit logs.
Reliability	Timestamped data, validation status, source provenance and graceful handling of missing data.
Explainability	Outputs understandable to non-technical officials.
Accessibility	Usable on desktop and mobile, including low-bandwidth workflows where feasible.

14. Security, Privacy & Governance

- Use aggregated population information wherever individual-level detail is not needed.
- Apply role-based access control and least-privilege permissions.
- Encrypt data in transit and at rest.
- Maintain audit logs for sensitive actions and assessment changes.
- Record data source, timestamp, validation status and model version.
- Provide human review / override workflow for critical assessments.
- Make uncertainty visible and avoid presenting model output as an absolute statement of safety.

15. MVP Scope

Recommended MVP: one pilot geography, 2-3 hazards, habitation-level assessment, shelter capacity, relocation priority and an interactive dashboard.

MVP feature	Included
Interactive GIS map	Yes
Habitation database	Yes
Flood / landslide risk layer	Yes
Population exposure	Yes
Vulnerability score	Yes
Risk score	Yes
Shelter capacity	Yes
Capacity deficit	Yes
Relocation priority	Yes
Candidate shelters	Yes
Basic route visualization	Yes
Authority dashboard	Yes

16. Success Metrics

Metric	What to measure
Risk identification accuracy	Agreement with validated high-risk locations in the pilot
Exposure estimation error	Difference between estimated and verified exposed population
Capacity accuracy	Difference between recorded and verified usable capacity
Decision-support time	Time for an officer to identify risk -> exposure -> shelter -> deficit -> options
System performance	Map/API response time and availability
Field verification rate	Share of critical assessments verified by authorized personnel

17. Risks & Mitigations

Risk	Potential effect	Mitigation
Poor / stale data	Incorrect assessment	Quality checks, freshness indicators, provenance
False positives	Unnecessary attention / resource use	Human review, calibrated thresholds, validation
False negatives	Missed vulnerable area	Conservative safeguards, field verification, uncertainty display
Real-time data gaps	Delayed risk updates	Fallback data sources and explicit stale-data flags
Capacity misreporting	Poor relocation planning	Occupancy verification and field updates
Over-reliance on AI	Unsafe decisions	Decision-support framing, explanations

18. Future Enhancements

- Predictive “what-if” simulations for changing rainfall, water levels or hazard conditions.
- Digital-twin representation of the monitored region.
- IoT integration with water-level, rainfall and soil sensors.
- Satellite/drone computer vision for flood expansion, road blockage and damage assessment.
- Verified community reporting workflows.
- Resource allocation support for rescue teams, ambulances, food, water and medical supplies.

19. Example Output

HABITATION: Village ABC

Population: 4,280 | Exposed: 3,750

Flood: HIGH | Landslide: LOW | Erosion: HIGH

Overall assessed risk: 84/100 | Zone: RED

Available shelter capacity: 2,300 | Capacity deficit: 1,450

Relocation priority: CRITICAL

Key factors: high flood exposure, high vulnerable population, insufficient shelter capacity, poor accessibility

Status: Human authority review required

20. Development Roadmap

Phase	Deliverable
1. Data + GIS	Habitation, population, roads, rivers, hazards, shelters and map layers
2. Risk Engine	Hazard, exposure, vulnerability and combined risk scoring
3. Capacity Engine	Occupancy, usable capacity and capacity-deficit calculations
4. Relocation Engine	Priority scoring and candidate safer destinations
5. Dashboard	Interactive authority dashboard and habitation details
6. AI Enhancement	Predictive models, anomaly detection and explainability
7. Field Validation	Mobile reporting and comparison with ground observations

**PRODUCT VALUE: Where is the danger? -> Who is exposed? -> Can available facilities absorb them?
-> Which cases require urgent review? -> What safer capacity options exist?**